

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Experiment 2: Binary Classification using Naïve Bayes and K-Nearest Neighbors

1. Aim and Objective

To implement Naïve Bayes and K-Nearest Neighbors (KNN) classifiers for a binary classification problem, evaluate them using multiple performance metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

2. Dataset Description

The Spambase dataset is a benchmark binary classification dataset used to identify spam emails. It consists of numerical features extracted from email content and a binary class label indicating spam or non-spam.

Dataset Reference:

- Kaggle – Spambase Dataset

3. Preprocessing Steps

- Loaded the dataset using Pandas
- Separated features and target labels
- Performed stratified train–test split
- Applied feature scaling using StandardScaler

4. Implementation Details

- Implemented Gaussian, Multinomial, and Bernoulli Naïve Bayes classifiers
- Implemented baseline KNN classifier
- Tuned KNN hyperparameters using GridSearchCV and RandomizedSearchCV
- Compared KDTree and BallTree neighbor search algorithms

5. Visualizations

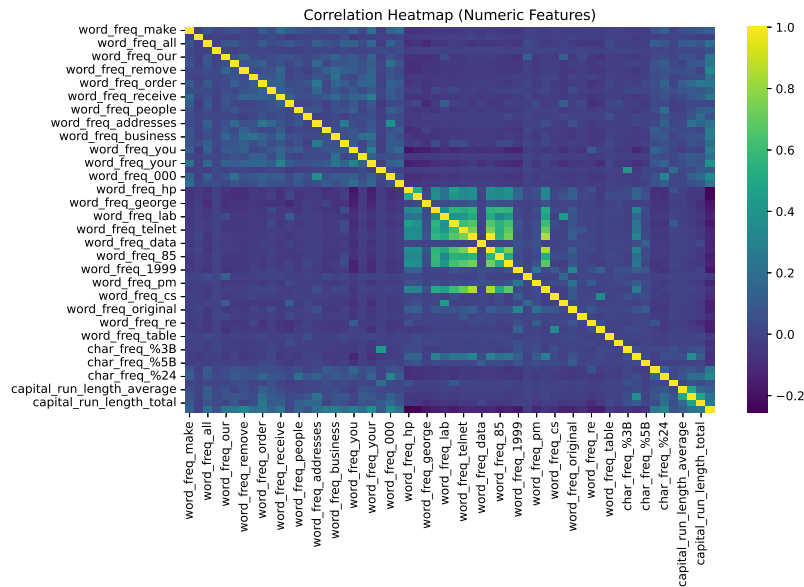


Figure 1: Heatmap showing correlation of elements

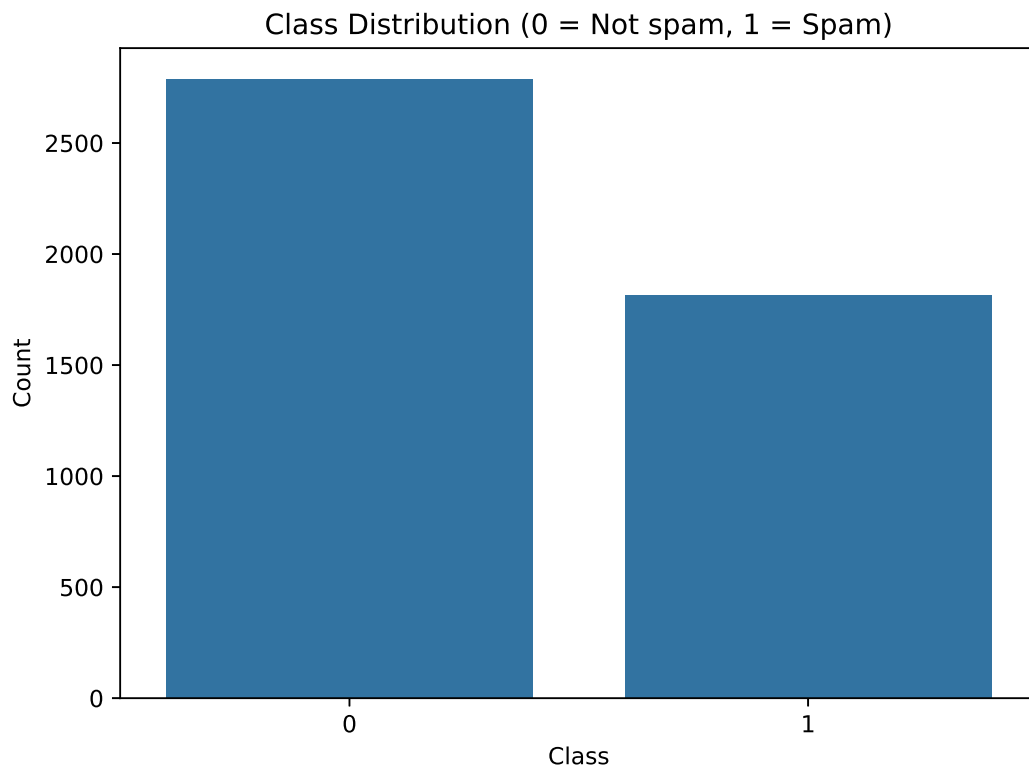


Figure 2: Class Distribution plot



Figure 3: Histogram plots showing feature distribution in numerical columns

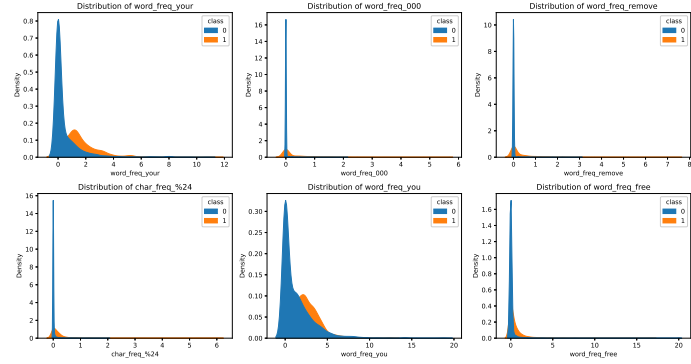


Figure 4: kde plots for most relevant features

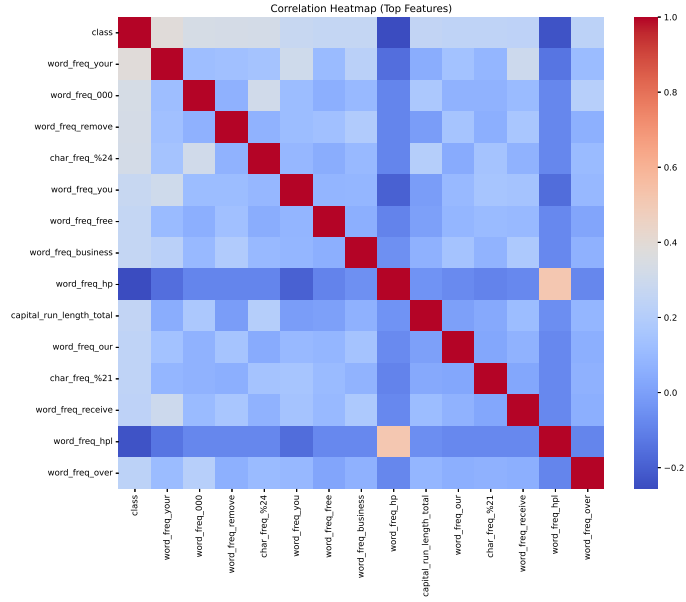


Figure 5: Correlation heatmap of top features

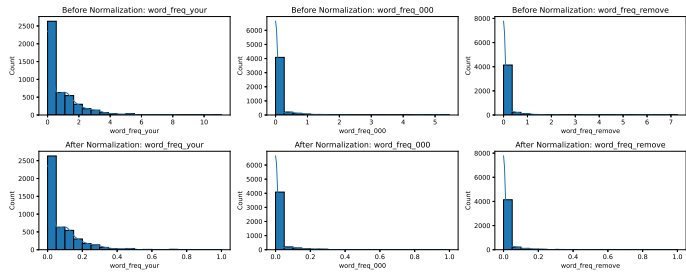


Figure 6: Before/After normalisation of elements

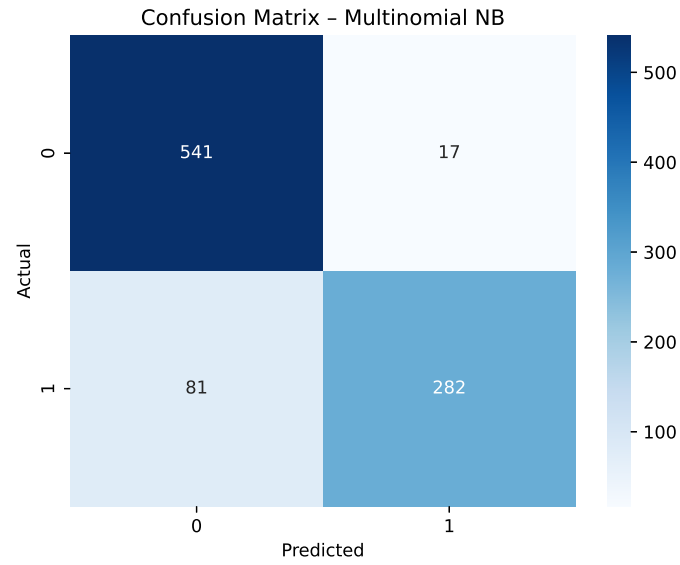


Figure 7: Confusion Matrix for Multinomial Naïve Bayes

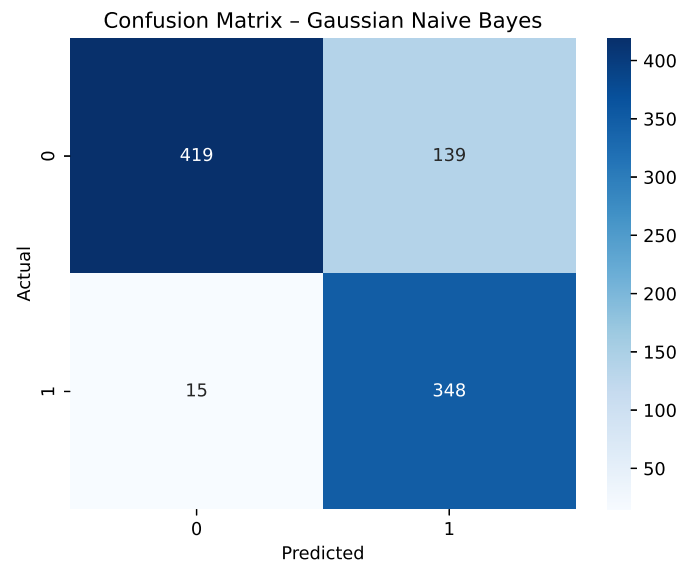


Figure 8: Confusion Matrix for Gaussian Naïve Bayes

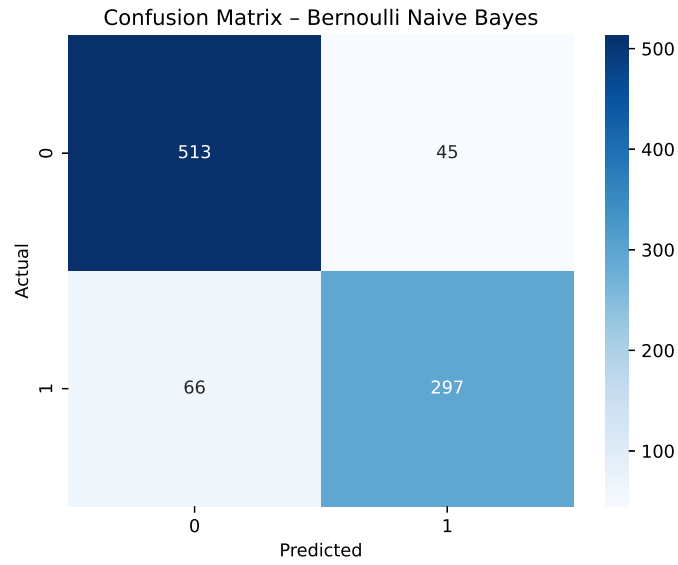


Figure 9: Confusion Matrix for Bernoulli Naïve Bayes

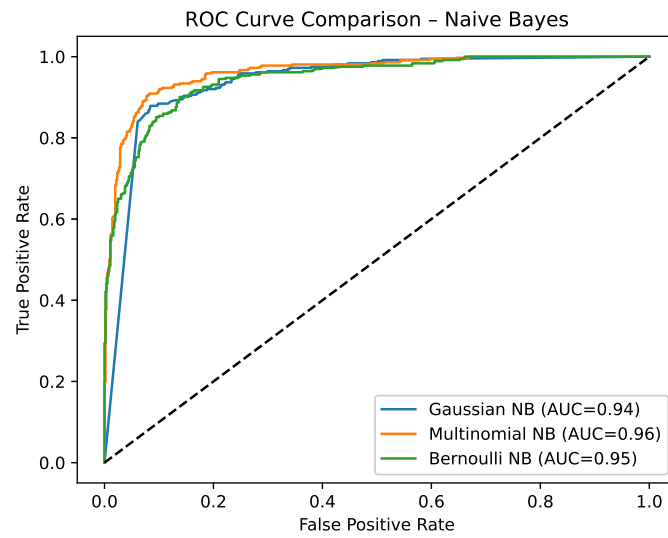


Figure 10: ROC curves of Multinomial, Naive Bayes, and Bernoulli

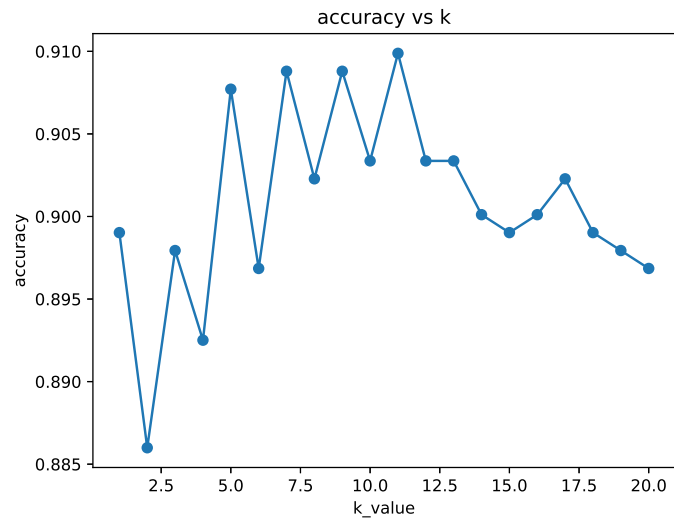


Figure 11: Accuracy vs K in KNN

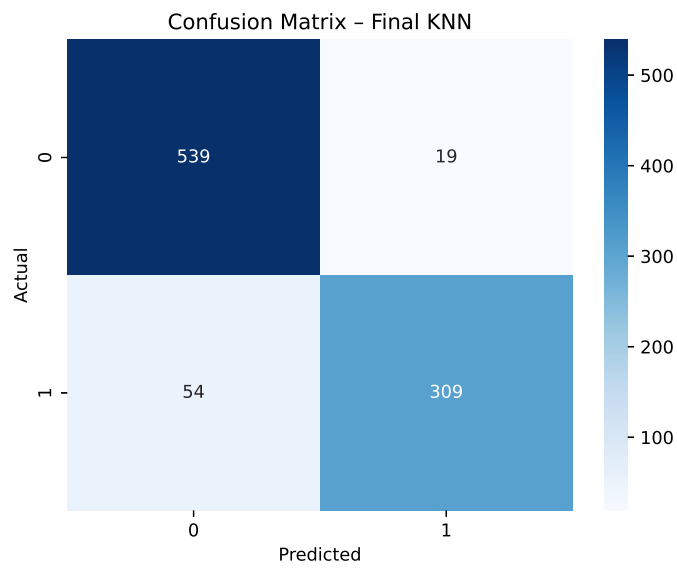


Figure 12: Confusion Matrix for Final KNN Model

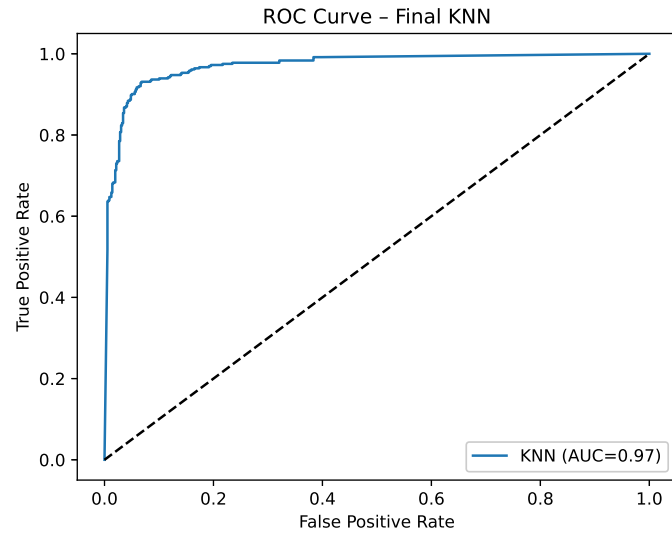


Figure 13: ROC curve of Final KNN value

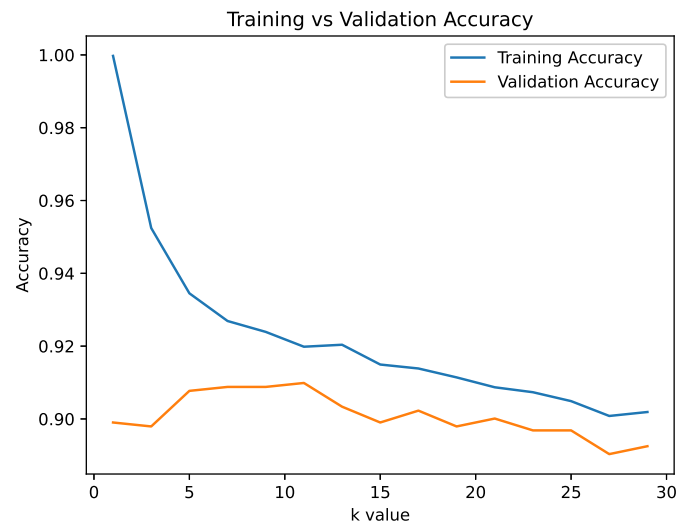


Figure 14: Training vs Validation Accuracy

6. Performance Tables

Table 1: Naïve Bayes Performance Metrics

Metric	Gaussian NB	Multinomial NB	Bernoulli NB
Accuracy	0.83	0.89	0.88
Precision	0.84	0.91	0.88
Recall	0.85	0.87	0.87
F1 Score	0.83	0.88	0.87
Specificity	0.7509	0.9695	0.9193
Training Time (s)	0.0091	0.0277	0.0152

Table 2: KNN Hyperparameter Tuning

Search Method	Best Parameters	Best CV Accuracy
Grid Search	k=9, distance, KDTree	0.9252
Randomized Search	k=6, distance, BallTree	0.9209

Table 3: KNN Performance using KDTree

Metric	Value
Optimal k	9
Accuracy	0.9207
Precision	0.9420
Recall	0.8512
F1 Score	0.8943
Training Time (s)	0.0038
Prediction Time (s)	0.0484

Table 4: KNN Performance using BallTree

Metric	Value
Optimal k	6
Accuracy	0.9218
Precision	0.9266
Recall	0.8705
F1 Score	0.8977
Training Time (s)	0.0038
Prediction Time (s)	0.0484

Table 5: Comparison of Neighbor Search Algorithms

Criterion	KDTree	BallTree
Accuracy	0.9207	0.9218
Training Time (s)	0.0038	0.0038
Prediction Time (s)	0.0484	0.1619
Memory Usage	Low / Medium	Medium / High

7. Overfitting and Underfitting Analysis

- Small values of k resulted in overfitting
- Large values of k caused underfitting
- Hyperparameter tuning improved generalization

8. Bias–Variance Analysis

- Naïve Bayes shows higher bias due to feature independence assumption
- KNN exhibits higher variance for small k values
- Hyperparameter tuning balances bias–variance trade-off

9. Observations and Conclusion

Naïve Bayes classifiers were computationally efficient, while tuned KNN models achieved higher accuracy. BallTree demonstrated better computational efficiency compared to KDTree. Proper preprocessing, visualization, and hyperparameter tuning significantly improved model performance and generalization.

References

- Scikit-learn – Naïve Bayes Documentation
- Scikit-learn – KNN Documentation
- Scikit-learn – Hyperparameter Optimization
- Kaggle – Spambase Dataset